Feature-Based Machine Unlearning for Vertical Federated Learning in IoT Networks

Zijie Pan, Zuobin Ying, Yajie Wang, Chuan Zhang, Weiting Zhang, Wanlei Zhou, and Liehuang Zhu

***Abstract*—In the era of the Internet of Things (IoT), managing the deluge of data generated by distributed devices presents unique challenges, particularly concerning privacy and the efficient use of computational resources. Vertical Federated Learning (VFL) offers a promising avenue for collaborative machine learning without centralizing data, thereby addressing privacy concerns inherent in traditional approaches. However, as data privacy laws and personal data deletion requests become more prevalent, the necessity for effective machine unlearning strategies within VFL frameworks grows increasingly important. To this end, this paper introduces a novel approach to feature-based machine unlearning tailored specifically for VFL systems in IoT networks. Our methodology enables the selective removal of data influence from trained models without the need for full retraining, thus preserving model utility while ensuring compliance with privacy requirements. By integrating a combination of feature relevance measuring techniques and efficient communication protocols, our solution minimizes the data footprint on network nodes, reduces bandwidth consumption, and maintains the integrity and performance of the learning models. To the best of our knowledge, our proposed framework represents the first practical approach to enable machine unlearning within vertical federated learning environments. We demonstrate the effectiveness of our approach through rigorous evaluation using several IoT datasets, highlighting significant improvements in unlearning efficiency and model robustness compared to existing techniques. Our work not only furthers the development of sustainable and compliant machine learning models in IoT but also sets a foundational framework for future research in secure and efficient data management within federated environments.**

***Index Terms*—Machine learning, privacy protection, federated learning.**

I. Introduction

THE advent of the Internet of Things (IoT) represents a new paradigm in how data is collected, processed, and utilized across a myriad of sectors. The proliferation of IoT devices — from personal health trackers and smart appliances to industrial sensors and urban traffic monitors — has enabled a continuous stream of data capturing intricate details of human behavior, environmental conditions, and machine performance. This vast and growing network of interconnected devices offers the potential to drive innovation and efficiency, providing personalized services, optimizing resource management, and enhancing decision-making processes. However, the deployment and operation of these devices raise complex challenges, particularly in data privacy, security, and sustainable management.

Vertical Federated Learning (VFL) has surfaced as a compelling approach to harness the potential of this data while mitigating the privacy concerns inherent in traditional centralized data processing systems. VFL allows different stakeholders, each holding a vertical slice of data pertaining to the same set of entities, to collaboratively train machine learning models. Each participant improves the model using their own data, updates model parameters locally, and shares only these updates — not the raw data — across the network. This methodology effectively prevents the exposure of sensitive data, ensuring that privacy is maintained while still leveraging collective insights to build robust predictive models.

Despite its advantages, VFL is not devoid of challenges. One of the most significant issues is related to data retention and the privacy of information encoded in the model parameters. The model inherently learns and retains sensitive information across its training process, potentially leading to privacy leaks if not handled correctly. Furthermore, participants in a VFL system may at any time decide, or be legally compelled, to remove their data from the model. This necessity can arise from regulatory requirements such as the GDPR, which includes stipulations for data rectification and the right to erasure — often colloquially referred to as ”the right to be forgotten”.

Addressing these complexities, this paper proposes a novel Feature-based Machine Unlearning methodology designed

Received 12 June 2024; revised 22 October 2024; accepted 2 January 2025. Date of publication 13 February 2025; date of current version 7 May 2025. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62402040, in part by the Postdoctoral Fellowship Program of CPSF under Grant GZB20230938, in part by the China Postdoctoral Science Foundation under Grant 2024T171132 and Grant 2023M740246, in part by FDCT under its General R&D Subsidy Program Fund under Grant 0038/2022/A, Macau, S.A.R., China, in part by NSFC-FDCT under its Joint Scientific Research Project Fund under Grant 0051/2022/AFJ, Macau, S.A.R., China, in part by Guangzhou University Research Project under Grant RQ2021013, in part by the Open Foundation of Key Laboratory of Cyberspace Security, Ministry of Education of China under Grant KLCS20240209, and in part by the National Natural Science Foundation of China under Grant 62472032. Recommended for acceptance by C. Assi. *(Corresponding author: Yajie Wang.)*

Zijie Pan, Zuobin Ying, and Wanlei Zhou are with the Faculty of Data Science, City University of Macau, Macau 314100, China (e-mail: zijiepan20@gmail.com; zbying@cityu.edu.mo; wlzhou@cityu.edu.mo).

Yajie Wang and Liehuang Zhu are with the School of Cyberspace Science and Technology, Beijing Institute of Technology, Beijing 100811, China (e-mail: wangyajie19@bit.edu.cn; liehuangz@bit.edu.cn).

Chuan Zhang is with the School of Cyberspace Science and Technology, Beijing Institute of Technology, Beijing 100811, China, and also with Key Laboratory of Cyberspace Security, Ministry of Education of China, China (e-mail: chuanz@bit.edu.cn).

Weiting Zhang is with the School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, China (e-mail: wtzhang@bjtu.edu.cn).

Digital Object Identifier 10.1109/TMC.2025.3530529

1536-1233 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

specifically for VFL systems in IoT contexts. Our approach aims to refine the unlearning process, enabling the selective removal of data influence from a model without necessitating full model retraining. This process not only adheres to privacy regulations but also preserves the model's utility and accuracy. Our proposal employs a combination of privacy-preserving mechanisms and sophisticated algorithmic techniques. Key strategies include the use of feature relevance scoring to identify and prioritize data features for removal, communication-efficient protocols to minimize data exchange, and learning rate adjustment as well as verification schemes. To the best of our knowledge, our proposal is the first practical framework to allow machine unlearning in vertical federated learning.

We face the following challenges in designing our unlearning framework. First, Vertical federated learning inherently involves the distribution of different features of datasets across various stakeholders without exchanging actual data points. To safeguard this fragmented data during the unlearning process, it is imperative to develop sophisticated privacy-preserving mechanisms. These should offer granular control over data features, enabling selective erasure that complies with stringent privacy standards without compromising the dataset's integrity. Second, given the limited bandwidth and diverse capabilities of IoT devices, optimizing communication is crucial. Our focus is on formulating communication protocols that minimize the exchange of data during unlearning. This involves leveraging advanced data compression techniques and efficient privacy-preserving methods that reduce overhead and facilitate timely data synchronization across the network. Third, ensuring the accuracy of the unlearning process is paramount. Our proposal must completely eliminate the influence of forgotten features while maintaining the overall model's performance. To achieve this, we explore algorithmic solutions capable of precisely identifying and nullifying the contributions of specific data features, supplemented by audit mechanisms that provide verifiable proof of successful unlearning.

To thoroughly evaluate the effectiveness of our FMU approach, extensive experiments were conducted using diverse and realistic IoT datasets. These datasets encompass various domains such as healthcare, where data sensitivity is paramount; urban planning, involving large-scale sensor data; and industrial IoT, which includes operational data from manufacturing processes. Our experiments are designed to simulate real-world scenarios where data may need to be selectively unlearned without compromising the overall functionality and accuracy of the model. These empirical evaluations provide insights into several key aspects of FMU, including the computational efficiency of unlearning processes, the integrity of data privacy after unlearning, and the impact on model performance. The results clearly demonstrate that FMu can rapidly and effectively dissociate data features from the model, ensuring compliance with privacy requirements while minimally affecting the predictive performance.

To summarize, this paper makes the following contributions:

- We present a new method for efficiently removing data features from a model in vertical federated learning setups within IoT environments. This process respects privacy regulations and maintains model accuracy without needing a complete retraining, addressing key challenges in data unlearning.
- The methodology leverages advanced cryptographic techniques combined with machine learning algorithms to ensure that the unlearning process is not only efficient but also secure. This is particularly important in IoT contexts where data sensitivity is high and the integrity of data handling is crucial.
- The effectiveness of the FMU approach is demonstrated through rigorous experiments using various IoT datasets, such as healthcare, urban planning, and industrial data. These experiments assess the computational efficiency of the unlearning process, the integrity of data privacy post-unlearning, and the impact on model performance.

The paper is organized as follows. Following this introduction, Section II reviews the relevant literature and discuses related work. Section III introduces preliminaries of this paper. Section IV presents our proposed methodology of feature based machine unlearning in federated learning. Section VI gives our designs for unlearning efficiency optimization. Section VII describes our experimental setup and presents empirical results. Finally, Section VII concludes our work.

II. Related Work

A. Vertical Federated Learning

Vertical Federated Learning (VFL) is a highly specialized form of federated learning that caters to scenarios in which multiple parties, each holding distinct data features for a common set of samples, collaborate to develop machine learning models. VFL stands distinct from its counterpart, Horizontal Federated Learning, in that it addresses the challenge of model training where the parties' data sets share the same sample space but differ in feature space. This method is particularly applicable in industries like healthcare, financial services, and any cross-organizational collaboration where sharing raw data is infeasible due to privacy laws, confidentiality agreements, or competitive boundaries [1], [2].

Initially conceptualized to tackle the stringent demands for privacy and data security in sensitive sectors, VFL has grown from theoretical frameworks to practical implementations [3], [4]. Early works in this field concentrated on creating secure environments where data could be used to collectively enhance model accuracy without ever leaving the owner's domain [5], [6]. This included employing advanced cryptographic methods such as homomorphic encryption, which allows computations to be performed on encrypted data, and secure multi-party computation, which facilitates a collaborative computation while keeping each party's input private [7], [8], [9]. Key foundational research by Bonawitz et al. [10] and Gascón et al. [11] broke ground by providing protocols that addressed how encrypted data could be used in federated models without compromising security.

As VFL models began to handle increasingly complex and voluminous data, the focus shifted towards optimizing the efficiency of algorithms to manage the scale and diversity of data without overwhelming network or computational resources [12],

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

[13]. This was particularly crucial in scenarios like IoT deployments, where devices at the edge are often limited by computational power and bandwidth [14], [15]. Dimlioglu et al. [16] have made significant contributions by proposing optimized federated learning algorithms that are tailored to reduce the amount of data transferred during the learning process, thus preserving bandwidth while maintaining the integrity of the model updates.

The adaptability of VFL to a range of machine learning models has opened up numerous possibilities for its application [17], [18], [19]. From linear models used in risk assessment to complex deep learning models for automated diagnosis in healthcare, VFL has proven capable of supporting diverse analytical tasks while safeguarding user privacy [20], [21]. Research led by Peter et al. [22] illustrates the integration of deep learning techniques into VFL, showcasing how neural networks can be trained using vertically partitioned data without exposing sensitive information. The continual evolution of VFL has seen the introduction of novel methodologies such as model compression techniques, which reduce the size of the model to be shared among the parties, and the integration of differential privacy to provide probabilistic guarantees that the models do not leak individual data points [23], [24]. These innovations enhance the practicality and security of VFL [25], [26].

Despite its advancements, VFL still faces numerous challenges. Key among these is the need to handle non-IID data distributions effectively, where data across different parties can exhibit varied statistical characteristics, potentially leading to biased or underperforming models [27]. Moreover, as the regulatory landscape around data privacy continues to evolve, VFL systems must adapt to meet these changing requirements [28]. Future research in VFL is likely to explore more robust algorithms for dynamic and adaptive learning, enhanced methods for secure data sharing in real-time, and the development of standardized protocols that could facilitate broader adoption across industries.

TABLE I
LIST OF FREQUENTLY USED NOTATIONS OF THIS PAPER

Notation	Description
θ	The set of parameters of the machine learning model.
η	The learning rate used in gradient descent methods.
X_{del}, Y_{del}	The features and labels of the data points to be deleted from the model.
$L(\theta; X, Y)$	The loss function of the model, where X and Y are the features and labels, respectively.
$\nabla_\theta L$	The gradient of the loss function with respect to the model parameters θ.
$\mathcal{N}(0, \sigma^2 I)$	Gaussian noise with mean 0 and variance σ^2, added to ensure differential privacy during the unlearning process.
α	A scaling factor used in approximate unlearning to adjust the influence of the subtraction data.
λ_i	Weights assigned to each sample in the subtraction dataset, reflecting its importance in the unlearning process.
D_i^{sub}	The subtraction dataset designed to mimic the negative impact of the deleted data.
H_θ^{-1}	The inverse Hessian matrix of the loss function, used in influence function optimization.
γ	Adjustment factor in influence function optimization, modulating the update intensity.
$V(\theta, \theta')$	A verification function that measures the fidelity of unlearning by comparing the model parameters before (θ) and after (θ') unlearning.
ϵ	A threshold value used in verification to ensure minimal impact from unlearned data.

B. Machine Unlearning

Machine unlearning has become increasingly significant in machine learning, addressing the issue of removing data effectively from models in compliance with legal standards such as the General Data Protection Regulation (GDPR). The concept of machine unlearning seeks to negate the influence of specific data from a trained model, ensuring the model behaves as if it had never encountered the removed data [29], [30]. This capability is not only crucial for privacy and compliance but also for adapting models in dynamic environments where data relevance may change over time [31], [32]. Initial frameworks for machine unlearning, as introduced by Cao and Yang [33], centered on decremental learning approaches. These methods systematically reduce the impact of data points on the model's learned parameters through direct modification techniques, although they often require significant computational resources [34], [35].

Efficiency in unlearning has been a focal point of subsequent research. The SISA [36] methodology developed by Bourtoule et al. is one such innovation. By partitioning the data and training isolated models on these partitions, the SISA method reduces the scope of retraining when unlearning is requested, thereby saving on computational costs. This method demonstrates considerable improvements in operational efficiency, especially when frequent data removal is anticipated [37].

Further advancements have explored hybrid approaches that combine traditional and novel unlearning methods to optimize both performance and efficiency [38], [39], [40]. For instance, techniques have been developed to approximate unlearning by selectively pruning and adjusting neurons in neural network models [41]. These approaches seek to balance the trade-offs between thorough unlearning and the computational overhead associated with more rigorous methods [42].

The verification of unlearning effectiveness has also emerged as a critical research area. Ensuring that unlearned data truly has no residual influence on the model is pivotal for trust and compliance [43]. Techniques involving retraining verification, statistical tests, and performance comparisons pre- and post-unlearning are used to validate the integrity of the unlearning process [44]. These verification methods are crucial for applications in sensitive fields such as healthcare and finance, where inaccuracies in data handling could have severe consequences [45].

Moreover, the scalability challenges of machine unlearning are particularly pronounced in scenarios involving large datasets or complex federated learning environments [46]. Recent studies have addressed these challenges by developing scalable unlearning algorithms that leverage distributed computing resources to parallelize the unlearning process [47]. Such scalability is vital for enabling efficient unlearning across vast networks of data and models, as typically seen in IoT systems.

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

III. Preliminaries

A. Vertical Federated Learning

VFL is specifically designed for collaborative machine learning scenarios where participants, each holding disjoint subsets of features for the same samples, work together to train a model without sharing sensitive data directly. This approach safeguards privacy while enabling collective intelligence. The optimization objective in VFL includes a compound loss function which combines the weighted local losses and a regularization term that models complexity:

$$C(\theta) = \sum_{i=1}^{n} \lambda_i \left(L_i(\theta; D_i) + \rho R(\theta) \right), \tag{1}$$

where $C(\theta)$ is the global cost function, λ_i weights the importance of each node's data, L_i are the local loss functions, D_i is the data held by participant i, and ρ controls the strength of regularization. A typical VFL process is proceed as follows.

Step 1: Data Alignment. Ensuring accurate alignment is critical as it allows participants to collaboratively train on the same instance set. Formally:

$$\text{Keys} = \bigcap_{i=1}^{n} \text{Keys}_i, \tag{2}$$

where Keys_i represents the unique identifiers from the dataset of participant i.

Step 2: Local Computation: Nodes calculate gradients using their local datasets. Each gradient reflects the partial derivative of the loss function with respect to the model parameters, considering only the local features:

$$g_i = \nabla_\theta L(\theta; X_i, Y_i), \tag{3}$$

where L is the loss function applied over the data X_i, Y_i. Using stochastic gradient descent (SGD), gradients are computed over mini-batches by clients locally:

$$g_{i,b} = \frac{1}{|B_i|} \sum_{b \in B_i} \nabla_\theta l(\theta; x_b, y_b), \tag{4}$$

where B_i represents a mini-batch of samples from participant i's data, and l is the loss per data point. In particular, each gradient may also be regularized to prevent overfitting and to improve generalization across the federation.

$$g_i^{reg} = g_i + \rho \nabla_\theta R(\theta) \tag{5}$$

where ρ is the regularization parameter, and $R(\theta)$ is a regularization term, typically a norm penalty on the weights.

Step 3: Gradient Aggregation: Gradients are aggregated across all participants to update the global model parameters. This aggregation often includes normalization factors to account for different data sizes or variances across nodes.

$$\theta \leftarrow \theta - \eta \left(\sum_{i=1}^{n} \frac{\lambda_i}{Z_i} g_i^{reg} \right), \tag{6}$$

where η is the learning rate, λ_i are the weights, and Z_i are normalization factors for the gradients from each participant. In practice, a momentum would be incorporated into the update rule to enhance convergence speeds by integrating velocity components from previous gradients.

$$v^{(t)} = \beta v^{(t-1)} + \sum_{i=1}^{n} \lambda_i g_i^{(t)}, \tag{7}$$

$$\theta^{(t+1)} = \theta^{(t)} - \eta v^{(t)}, \tag{8}$$

where $v^{(t)}$ is the velocity at time t, and β is the momentum factor.

B. Unlearning in Neural Networks

Implementing machine unlearning processes ensures that models remain compliant with legal standards and maintain the trust of their users by respecting individual privacy rights. Based on practical needs, current solutions can be characterized into Exact Unlearning and Approximate Unlearning. Consider a dataset $\mathcal{D}$, a to be unlearned dataset $\mathcal{D}_u$, and the remaining dataset $\mathcal{D}_r = \mathcal{D}/\mathcal{D}_u$. Machine unlearning is defined as follows.

Machine Unlearning: Machine unlearning refers to the process of removing the influence of specific data points from a trained machine learning model, ensuring that the model behaves as if the data had never been seen. This procedure is crucial in scenarios where data must be deleted to comply with privacy regulations such as the General Data Protection Regulation (GDPR). The goal is to modify the model parameters such that the contribution of the removed data is eliminated, while preserving the model's utility for the remaining data.

Exact unlearning is often used to achieve this by retraining the model without the deleted data. For instance, if the data to be unlearned is represented by (X_{del}, Y_{del}), the retrained model parameters θ_{dp} are computed as:

$$\theta_{dp} = \theta - \eta \left(\nabla_\theta L(\theta; X_{del}, Y_{del}) + \mathcal{N}(0, \sigma^2 I) \right), \tag{9}$$

where θ are the current model parameters, η is the learning rate, and $\mathcal{N}(0, \sigma^2 I)$ introduces Gaussian noise to ensure differential privacy. The term $\nabla_\theta L(\theta; X_{del}, Y_{del})$ represents the gradient of the loss function with respect to the data (X_{del}, Y_{del}) to be removed, and the noise term obscures any trace of the data's removal, ensuring privacy.

Approximate Unlearning: Approximate unlearning is a more computationally efficient method that adjusts the model to minimize the influence of the data without full retraining. Instead of fully retraining the model, approximate unlearning incrementally updates the model parameters based on sensitivity analysis or influence functions. This approach offers a trade-off between computational cost and the precision of data removal.

In approximate unlearning, sensitivity analysis is used to estimate how much the model parameters change when a specific dataset is removed. The approximate update to the model parameters, denoted as θ_{sens}, can be calculated as:

$$\theta_{sens} = \theta - \alpha \left(\sum_{i=1}^{n} \lambda_i \nabla_\theta L_i(\theta; D_i^{sub}) + \Delta\theta \right), \tag{10}$$

where α is a scaling factor, λ_i are weights for each subtraction dataset sample D_i^{sub} (which approximates the influence of the

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

removed data), and $\Delta\theta$ represents a fine-tuning adjustment for stability.

A common technique for approximate unlearning involves using influence functions to estimate the effect of individual data points on the learned parameters. The update is given by:

$$\theta_{if} = \theta - \gamma \left(H^{-1}\theta \nabla \theta L(\theta; x_{del}, y_{del}) \right), \tag{11}$$

where γ controls the strength of the update, $H^{-1}\theta$ is the inverse Hessian matrix of the loss function, and $(xdel, y_{del})$ are the specific data points being unlearned. This allows the model to efficiently "forget" the targeted data points.

Verification of Unlearning: Verifying that data has been effectively unlearned is an important aspect of the process. This involves ensuring that the model with the unlearned data behaves similarly to a model trained without that data. Verification can be formalized by comparing the output of the updated model θ_{new} with the original model θ:

$$V(\theta_{new}, \theta) \leq \epsilon, \quad V = |\mathbb{E}[f(x; \theta_{new})] - \mathbb{E}[f(x; \theta)]|, \tag{12}$$

where V is a verification function that measures the difference in the model's behavior before and after unlearning, and ϵ is a predefined threshold ensuring that the unlearned model meets the desired privacy requirements. Table I summarizes the frequently used notations in this paper.

IV. Methodology

This section outlines our proposed methodology for implementing Feature-based Machine Unlearning in VFL systems within IoT networks. Our approach centers around a system model that utilizes a multi-layered distributed architecture, allowing IoT devices to process data locally and only share model updates across the network, thus enhancing privacy and reducing bandwidth use. Then, we detail our the unlearning process, which is designed to selectively remove the influence of specific data features or instances without degrading the model's overall performance. This is achieved through a feature relevance scoring mechanism, which guides the weighted aggregation of updates and ensures the precise removal of data features.

Our framework is adaptable to domains like finance and healthcare, where privacy is critical. In finance, feature-based unlearning allows selective removal of sensitive data, ensuring compliance with regulations while maintaining model performance. In healthcare, differential privacy and robust unlearning secure patient data, enabling safe analysis without compromising privacy. Its efficient, scalable design makes the framework ideal for applications where data privacy must be balanced with accurate predictions.

A. System Model and Assumptions

The federated learning network in this paper is architecturally configured as a distributed multi-layered hierarchy where IoT devices are clustered based on geographic proximity and functional similarities. Each cluster C_i is overseen by a Cluster Head (CH), which acts as the central node for that cluster. The CH facilitates all intra-cluster communications and aggregates data for transmission to the Central Server (CS), which coordinates learning processes across the entire network.

The network topology is represented mathematically by a directed graph $G(V, E)$, where V denotes the set of clusters and E the set of edges representing communication links:

$$A = [a_{ij}]_{n\times n}, \quad a_{ij} = \begin{cases} 1 & \text{link from } i \text{ to } j \text{ exists,} \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

Here, n represents the total number of clusters.

Data Flow and Communication Protocol: Data collection within each cluster is initially processed locally to extract relevant features, reducing transmission overhead and preserving bandwidth. We presume the aggregation at the cluster head utilizes a dynamic weighting algorithm that accounts for the temporal variability and source reliability of data:

$$f_{\text{agg}}(x_1, x_2, \ldots, x_k) = \sum_{i=1}^{k} w_i(t) \cdot x_i,$$
$$\text{s.t. } w_i(t) = \frac{q_i(t) \cdot e^{-\alpha d_i}}{\sum_{j=1}^{k} q_j(t) \cdot e^{-\alpha d_j}}. \tag{14}$$

where x_i is the data from device i, $q_i(t)$ is the quality score at time t, d_i is the time delay in data reception, and α is a decay factor that reduces the weight of older, potentially outdated data.

The central server computes the global model update using a weighted aggregation of the data received from all clusters. The update is mathematically formulated as:

$$M_{t+1} = M_t + \eta \cdot \left(\frac{1}{n} \sum_{i=1}^{n} f_{\text{agg}}(x_{i1}, x_{i2}, \ldots, x_{ik_i}) \right), \tag{15}$$

where M_t is the global model at iteration t, η is the learning rate, and f_{agg} is the aggregated data from each cluster.

Following the aggregation, the updated model M_{t+1} is distributed back to each cluster:

$$M_{i,t+1} = M_{i,t} + \beta_i(t) \cdot (M_{t+1} - M_t), \tag{16}$$

where $\beta_i(t)$ is a scaling factor specific to each cluster, adjusted dynamically based on the cluster's current data accuracy and network conditions.

B. Unlearning

Central to our proposal is selective removal of data influences from a federated learning model in a vertical IoT setting. Each local model, M_i, is trained on vertically partitioned datasets, which means that each IoT device processes a unique subset of features. Our proposed method consists of several components. Fig. 1 gives a high level overview of our proposal.

Feature Relevance Scoring Mechanism: Given the diversity in data distribution and the significance of the features, we propose a feature relevance scoring mechanism:

$$\theta_i = \text{train}(D_i), \quad D_i = \{x_{ij} | j \in \text{features}\}, \tag{17}$$

$$\mathcal{R}(x_{ij}) = \frac{\text{var}(x_{ij})}{\sum_{k\in\text{features}} \text{var}(x_{ik})} \tag{18}$$

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

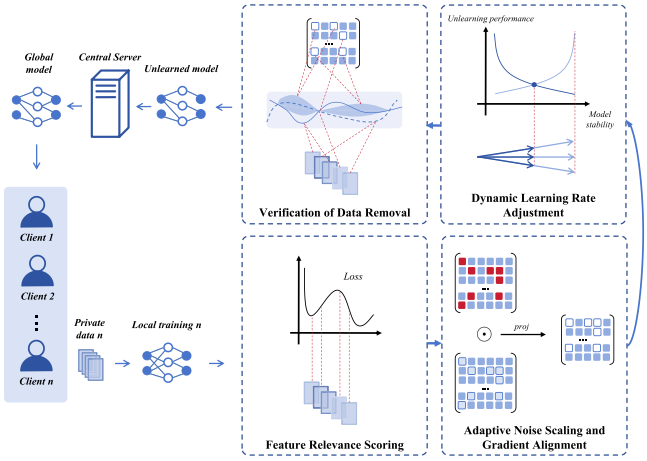


Fig. 1. A general pipeline of our proposal.

Here, $\text{train}(D_i)$ is the local training function and $\text{var}(x_{ij})$ denotes the variance of feature j in dataset i, $\mathcal{R}(.)$ is the feature relevance scoring function. On this basis, the relevance scores guide the weighted aggregation:

$$M_I = \sum_{i=1}^{N} w_i M_i, \quad w_i = \frac{\sum_{j\in\text{features}} \mathcal{R}(x_{ij})}{\sum_{i=1}^{N}\sum_{j\in\text{features}} \mathcal{R}(x_{ij})} \tag{19}$$

The unlearning phase of our algorithm is designed to ensure the precise removal of specific data features or instances, while simultaneously maintaining the integrity and performance of the federated learning model. To refine the calculation of the influence of the data points to be removed, we extend the influence function approach by incorporating integral operators that capture long-range dependencies within the data:

$$\Delta M_I = -\left(\int H_{LM}^{-1}(x,y)\cdot \sum_{f\in F} \nabla_f L(x; M_I, D)\, dx\right), \tag{20}$$

where $H_{LM}^{-1}(x,y)$ is the Green's function of the Hessian matrix, representing the inverse Hessian as a function of spatial variables x and y. This computation captures the global influence of features across the model's entire parameter space, ensuring a more accurate removal of their effects.

Adaptive Noise Scaling and Gradient Alignment: We extend the adaptive noise scaling mechanism by integrating feature-based relevance, underpinned by vertical federated data partitions, introducing dynamic perturbation elements. Let M denote the high-dimensional model tensor, and let $\Delta M(t)$ denote the time-indexed perturbation operator. Specifically, the perturbation process for tth training round adapts based on localized feature relevance and varying regulatory privacy gradients ϵ, parameterized as:

$$\Delta M(t) = \mathcal{N}(0, \sigma_k^2(t)\cdot \mathbf{I}) + \mathcal{N}(0, \tau_j^2(t)\cdot \mathbf{I}), \tag{21}$$

where $\sigma_k^2(t), \tau_j^2(t)$ are non-linear variance functions adhering to the exponential scaling, given by:

$$\sigma_k^2(t) = \sigma_0^2 \cdot \exp\left(\gamma_k \cdot \mathcal{S}_{\text{rel}}(t)\right), \tag{22}$$

$$\tau_j^2(t) = \tau_0^2 \cdot \exp\left(\lambda_j \cdot \frac{1}{\mathcal{S}_{\text{rel}}(t)+\epsilon}\right), \tag{23}$$

and $\mathcal{S}_{\text{rel}}(t)$ is related to the Lyapunov stability metrics and computed by measuring the norm of the gradient differences. This process encodes differential privacy under a temporally adaptive scaling mechanism.

Upon perturbation, we employ a constrained retraining process using orthogonality principles, described as:

$$M' = M - \eta\cdot \Delta M(t) + \int_{\mathbb{R}^d} \mathcal{G}_{\text{corr}}(x, M^*)\cdot dx, \tag{24}$$

where $\mathcal{G}_{\text{corr}}(x, M^*)$ denotes a corrective Green's function derived from the manifold curvature tensor of M^*, and ensures fine-grained recalibration through:

$$M'' = M' + \text{Proj}_{\text{span}(\mathcal{D}_{\text{retained}})}(M') = M' + \mathbf{Q}(\mathbf{Q}^\top\mathbf{Q})^{-1}\mathbf{Q}^\top M', \tag{25}$$

where $\mathbf{Q}\in\mathbb{R}^{d\times p}$ are orthonormal basis vectors satisfying orthogonal convergence criteria in $\mathcal{L}^2$-space, ensuring the recalibrated model is geometrically consistent with the preserved feature space.

Finally, to align the perturbation dynamics with federated learning paradigms, the noise added to the global $\mathcal{F}_{\text{noise}}$ is given by a multi-objective constraint system:

$$\mathcal{F}_{\text{noise}} = \arg\min_{\sigma,\tau} \int_{t=0}^{T} \left\|\Delta M(t)\right\|_{\text{F}}^2 + \lambda\|\mathcal{P}_{\text{reg}}(M')\|^2 dt, \tag{26}$$

where the Frobenius norm $\|\cdot\|_{\text{F}}$ tracks divergence and $\mathcal{P}_{\text{reg}}$ acts as a penalizing term to control overshooting during recalibration.

Dynamic Learning Rate Adjustment: The adaptive adjustment of the learning rate η is governed by a Lyapunov stability criterion to ensure the model's stability throughout the unlearning process:

$$\eta_t = \min\left(\eta_{t-1}\cdot\exp\left(-\frac{\|M_t - M_{t-1}\|}{\text{Lyap}(M_{t-1})}\right), \eta_{\max}\right), \tag{27}$$

where $\text{Lyap}(M_{t-1})$ is a Lyapunov function [48] measuring the stability of the model at iteration $t-1$. This criterion helps to dynamically adjust the learning rate, ensuring optimal stability and convergence during model updates.

V. Theoretical Analysis

A. Convergence Rate

We analyze the convergence of the global model in a VFL system under the influence of adaptive unlearning processes. Let M_t denote the global model parameters at iteration t. The goal is to derive a convergence bound for the global update process, incorporating the effects of adaptive noise scaling and dynamic gradient adjustments during unlearning.

Condition 1 (Smoothness): The loss function $L(\theta)$ is L-smooth, *i.e.*, there exists a constant $L>0$ such that:

$$\|\nabla L(\theta_1) - \nabla L(\theta_2)\| \le L\|\theta_1 - \theta_2\| \quad \forall\theta_1,\theta_2. \tag{28}$$

Condition 2 (Strong Convexity): The loss function $L(\theta)$ is μ-strongly convex, i.e., there exists a constant $\mu>0$ such that:

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

$$L(\theta_2) \geq L(\theta_1) + \nabla L(\theta_1)^\top (\theta_2 - \theta_1) + \frac{\mu}{2}\|\theta_2 - \theta_1\|^2. \quad (29)$$

Condition 3 (Bounded Variance): Let ξ_t represent the randomness introduced by the adaptive noise mechanism. The variance of the stochastic gradients is bounded by zero mean noise, *i.e.,* $\mathbb{E}[\delta_t] = 0$. Then:

$$\mathbb{E}[\|\nabla L(\theta, \xi) - \nabla L(\theta)\|^2] \leq \xi_t \leq \sigma_k^2(t) + \tau_j^2(t), \quad (30)$$

where $\nabla L(\theta, \xi)$ is the stochastic gradient at θ, and $\sigma_k^2(t) + \tau_j^2(t)$ is given by (22) and (23).

Condition 4 (Adaptive Learning Rate): Let η_t be the learning rate at iteration t, satisfying $0 < \eta_t < \frac{2}{L}$. We also assume η_t decreases over time to ensure convergence.

Given the above conditions and assumptions, the global model M_t converges to the optimal model M^* at a rate of:

$$\mathbb{E}[L(M_t) - L(M^*)] \leq \mathcal{O}\left(\frac{1}{t}\right). \quad (31)$$

B. Proof of Convergence Bound

We begin by consideing the expected squared distance to the optimal parameters:

$$\mathbb{E}\left[\|\theta_{t+1} - \theta^*\|^2\right]. \quad (32)$$

We aim to bound this term. Using the update rule: $\theta_{t+1} = \theta_t - \eta_t \nabla L(\theta_t, \xi_t)$ We get:

$$\theta_{t+1} - \theta^* = \theta_t - \theta^* - \eta_t \nabla L(\theta_t, \xi_t). \quad (33)$$

Compute the Squared Norm gives:

$$\begin{aligned}&\|\theta_{t+1} - \theta^2\| \\ &= \|\theta_t - \theta^2\| - 2\eta_t(\theta_t - \theta^*)^\top \nabla L(\theta_t, \xi_t) + \eta_t^2 \|\nabla L(\theta_t, \xi_t)\|^2.\end{aligned} \quad (34)$$

We then take the expectation conditioned on θ_t:

$$\begin{aligned}&\mathbb{E}\left[\ \|\theta_{t+1} - \theta^2\| \mid \theta_t\right] \\ &= \|\theta_t - \theta^2\| - 2\eta_t(\theta_t - \theta^*)^\top \mathbb{E}\left[\nabla L(\theta_t, \xi_t) \mid \theta_t\right] \\ &\quad + \eta_t^2 \mathbb{E}\left[\|\nabla L(\theta_t, \xi_t)\|^2 \mid \theta_t\right].\end{aligned} \quad (35)$$

Since $\mathbb{E}[\nabla L(\theta_t, \xi_t) \mid \theta_t] = \nabla L(\theta_t)$:

$$\begin{aligned}&\mathbb{E}\left[\ \|\theta_{t+1} - \theta^2\| \mid \theta_t\right] \\ &= \|\theta_t - \theta^2\| - 2\eta_t(\theta_t - \theta^*)^\top \nabla L(\theta_t) \\ &\quad + \eta_t^2 \mathbb{E}\left[\|\nabla L(\theta_t, \xi_t)\|^2 \mid \theta_t\right].\end{aligned} \quad (36)$$

Noticing $\mathbb{E}[\|\nabla L(\theta_t, \xi_t)\|^2]$ can be decomposed as:

$$\mathbb{E}\left[\|\nabla L(\theta_t, \xi_t)\|^2\right] = \|\nabla L(\theta_t)\|^2 + \mathbb{E}\left[\|\delta_t\|^2\right] + 2\nabla L(\theta_t)^\top \mathbb{E}\left[\delta_t\right], \quad (37)$$

$\mathbb{E}[\delta_t] = 0$, and $\mathbb{E}[\|\delta_t\|^2] \leq \sigma_k^2(t) + \tau_j^2(t)$, The expected squared distance becomes:

$$\begin{aligned}\mathbb{E}\left[\|\theta_{t+1} - \theta^2\| \mid \theta_t\right] = \ &\|\theta_t - \theta^2\| - 2\eta_t(\theta_t - \theta^*)^\top \nabla L(\theta_t) \\ &+ \eta_t^2 \left(\|\nabla L(\theta_t)\|^2 + \sigma_k^2(t) + \tau_j^2(t)\right).\end{aligned} \quad (38)$$

From the strong convexity and smoothness of $L(\theta)$, we have:

$$(\theta_t - \theta)^\top \nabla L(\theta_t) \geq \frac{\mu}{2}\|\theta_t - \theta^2\| + \left(L(\theta_t) - L(\theta^*)\right), \quad (39)$$

and

$$\|\nabla L(\theta_t)\|^2 \leq 2L\left(L(\theta_t) - L(\theta^*)\right). \quad (40)$$

Substitute these back into the expected squared distance:

$$\begin{aligned}&\mathbb{E}\left[\|\theta_{t+1} - \theta^2\| \mid \theta_t\right] \\ &\leq \|\theta_t - \theta^2\| - 2\eta_t\left(\frac{\mu}{2}\|\theta_t - \theta^2\| + L(\theta_t) - L(\theta^*)\right) \\ &\quad + \eta_t^2\left(2L\left(L(\theta_t) - L(\theta^*)\right) + \sigma_k^2(t) + \tau_j^2(t)\right).\end{aligned} \quad (41)$$

Simplify gives:

$$\begin{aligned}\mathbb{E}\left[\|\theta_{t+1} - \theta^2\| \mid \theta_t\right] \ \leq \ &(1 - \eta_t\mu)\|\theta_t - \theta^2\| \\ &- \left(2\eta_t - 2\eta_t^2 L\right)\left(L(\theta_t) - L(\theta^*)\right) \\ &+ \eta_t^2\left(\sigma_k^2(t) + \tau_j^2(t)\right).\end{aligned} \quad (42)$$

Since $0 < \eta_t < \frac{2}{L}, 1 - \eta_t L/2 > 0$, so $2\eta_t - 2\eta_t^2 L > 0$. Let $A_t = \mathbb{E}[\|\theta_t - \theta^*\|^2]$ and $V_t = \sigma_k^2(t) + \tau_j^2(t)$, (42) becomes

$$A_{t+1} \leq (1 - \eta_t\mu) A_t + \eta_t^2\left(\sigma_k^2(t) + \tau_j^2(t)\right), \quad (43)$$

which is a recurrence formulation.

Notably η_t decreases over time and $\sum_{t=1}^{\infty} \eta_t = \infty$, $\sum_{t=1}^{\infty} \eta_t^2 < \infty$. For simplicity, consider a constant learning rate $\eta_t = \eta$, where $0 < \eta < \frac{2}{L}$. The recurrence becomes: $A_{t+1} \leq (1 - \eta\mu)A_t + \eta^2 V_t$. Iterate the inequality, we have:

$$\begin{aligned}A_{t+1} &\leq (1 - \eta\mu)A_t + \eta^2 V_t \\ &\leq (1 - \eta\mu)^{t+1} A_0 + \eta^2 \sum_{k=0}^{t} (1 - \eta\mu)^{t-k} V_k.\end{aligned} \quad (44)$$

Since $0 < 1 - \eta\mu < 1$, we can bound the sum:

$$\sum_{k=0}^{t} (1 - \eta\mu)^{t-k} V_k \leq \sum_{k=0}^{t} V_k. \quad (45)$$

Assuming that $V_t \leq V_{\max}$ for all t, we have:

$$A_{t+1} \leq (1 - \eta\mu)^{t+1} A_0 + \eta^2 V_{\max} \cdot \frac{1 - (1 - \eta\mu)^{t+1}}{\eta\mu}. \quad (46)$$

This implies as $t \to \infty$, $(1 - \eta\mu)^{t+1} \to 0$, and we have:

$$\lim_{t \to \infty} A_{t+1} \leq \frac{\eta V_{\max}}{\mu}. \quad (47)$$

Thus,

$$\begin{aligned}\mathbb{E}\left[L(\theta_{t+1}) - L(\theta^*)\right] &\leq (1 - \eta\mu)^{t+1}\left(L(\theta_0) - L(\theta^*)\right) + \frac{\eta V_{\max}}{2} \\ &\leq \mathcal{O}\left(\frac{1}{e^t}\right) \leq \mathcal{O}\left(\frac{1}{t}\right).\end{aligned} \quad (48)$$

VI. Unlearning Optimizations

Gradient Optimization: To improve computational efficiency in gradient recalculation during unlearning, we combine

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

Stochastic Variance Reduced Gradient (SVRG) with kernel-based regularization:

$$\tilde{\nabla} M = \frac{1}{n}\sum_{i=1}^{n} \nabla l_i(M) + \sum_{i \in U} K(x_i, x)\nabla l_i(M),$$
$$\nabla M_t' = \nabla l_{i_t}(M_t) - \nabla l_{i_t}(\tilde{M}) + \tilde{\nabla} M, \quad (49)$$

where $K(x_i, x)$ is a kernel function that adjusts updates based on feature similarity, blending stochastic updates with full-gradient precision to accelerate convergence. After unlearning, we incorporate Group Lasso with non-convex penalties to ensure robust generalization:

$$L'(M) = L(M) + \lambda_{\text{GL}} \sum_{j=1}^{J} \|M^{(j)}\|_2 + \Phi(M), \quad (50)$$

where $\Phi(M)$ is a non-convex regularizer that promotes properties such as sparsity and low-rank structure, maintaining model integrity.

Quantum Annealing: We also employ a time-dependent Hamiltonian framework to guide optimization:

$$\mathcal{H}(t) = (1-t)\mathcal{H}_0 + t\mathcal{H}_P, \quad \mathcal{H}_P = -\sum_{i,j} J_{ij}\sigma_i^z\sigma_j^z - \sum_i h_i \sigma_i^x, \quad (51)$$

where $\mathcal{H}_0$ is the initial Hamiltonian, and $\mathcal{H}_P$ encodes the target optimization problem. The parameters J_{ij} and h_i control the system's evolution, while $\tilde{J}_{ij}(d_{ij})$ and $\tilde{h}_i(d_i)$ incorporate network-induced constraints reflecting data distribution and device interactions.

To manage distributed unlearning efficiently:

$$M'' = M - \sum_{j=1}^{J} \Gamma_j(\Delta M_j), \quad \Gamma_j = \alpha_j \cdot \exp(-\beta_j \|\Delta M_j\|), \quad (52)$$

where α_j and β_j are scaling factors adapted to computational capacity and data characteristics.

Parallel Processing and Batch Unlearning: On this basis, the update ΔM_j is averaged across nodes:

$$\Delta M_j = \frac{1}{N}\sum_{n=1}^{N} \Delta M_{j,n}, \quad \alpha_j = \frac{C_n}{C_{\max}}, \ \beta_j = \frac{S_{\max}}{S_j}, \quad (53)$$

where C_n denotes the computational capacity of node n, and S_j represents data sensitivity in batch j. The error metric is defined as:

$$\epsilon_j = \|\Delta M_j - \Gamma_j(\Delta M_j)\|, \quad (54)$$

and is used to guide the regularization process:

$$\min L(M'') = L(M) - \lambda \sum_{j=1}^{J} \|\Gamma_j(\Delta M_j)\|^2, \quad (55)$$

balancing the trade-off between unlearning precision and overall model accuracy.

VII. Experiments

This section outlines the experimental setup and results for our proposed method. Experiments employed different network architecture. Key metrics included model accuracy, forget rate, and the time required for unlearning and relearning. Results demonstrated the method's effectiveness in maintaining accuracy and achieving substantial forget rates efficiently.

A. Experimental Setup

Platform. We implement our proposed machine unlearning algorithm in Python and run experiments on a server with an Intel i9-13900k CPU and a NVIDIA RTX 4080 GPU, running Ubuntu 18.04 LTS. We train all neural network models using PyTorch.

Dataset: In the paper, we use 2 widely recognized public datasets to assess the efficacy of our unlearning algorithm. We use Bank [49], and Credit [50]. In particular, Bank consists of 41,188 records, 2 classes, and 20 features. Credit consists of 30,000 records, 2 classes, and 23 Features. We normalize the feature values in each data set within the range $[0, 1]$. These datasets were chosen due to their varied data types and real-world relevance in privacy-sensitive domains such as financial services.

VFL settings: In our experiments, we construct a VFL with 100 clients. In particular, the data point are distributed according to two settings: IID, and Non-IID. In IID setting, data points are evenly distributed among all participating client devices, ensuring each device receives a representative sample of the entire dataset. In Non-IID settings, data points are distributed among clients such that each participant receives a portion of the dataset that may differ significantly in its statistical properties from the portions held by other participants.

Model and Training Specifications: In our experiments, we designed two distinct model architectures to cater to the diverse computational capabilities and requirements of various client devices participating in the study. The first model, aimed at high-capacity client devices, was a deep convolutional neural network (CNN) tailored for complex feature extraction. This CNN architecture consisted of four convolutional layers with 64, 128, 256, and 512 filters respectively, each followed by a max-pooling layer to reduce dimensionality and capture the most salient features. Activation functions were ReLU to add non-linearity, and dropout layers were incorporated at a rate of 0.4 after each pooling layer to prevent overfitting. The model concluded with two fully connected layers of 1024 and 512 neurons, respectively, leading to a softmax output layer for classification. The optimizer used was Adam, with an initial learning rate of 0.0005, adjusted dynamically based on validation loss.

For lower-capacity client devices, a simpler and computationally less intensive model was employed. This model was a shallow neural network (SNN) consisting of two hidden layers with 32 and 16 neurons, ideal for devices with limited processing power. Each layer used a sigmoid activation function, which, while historically traditional, is suitable for smaller networks due to its smooth gradient behavior. A single dropout layer with a dropout rate of 0.3 was placed after the first hidden layer to aid

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

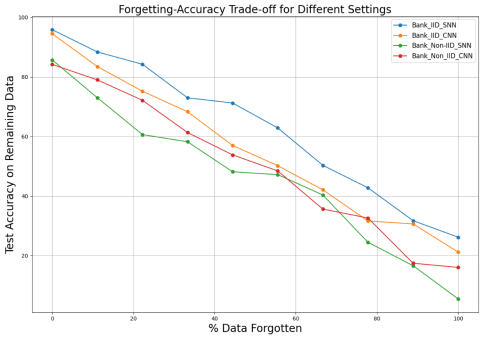


Fig. 2. Forgetting-accuracy trade-off for different settings of bank.

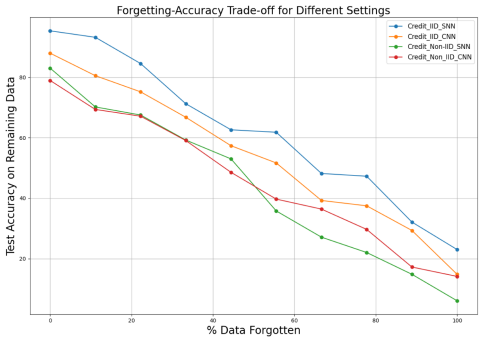


Fig. 3. Forgetting-accuracy trade-off for different settings of credit.

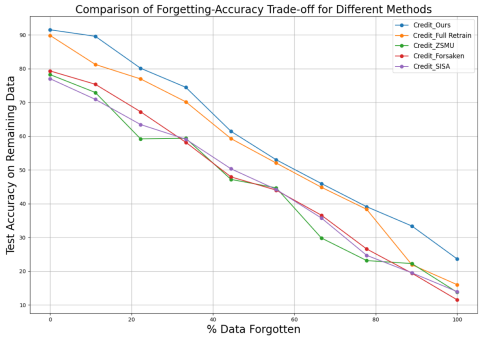


Fig. 4. Forgetting-accuracy trade-off of non_IID and CNN of bank.

in generalization without significantly increasing computational demands. The output layer was a softmax layer, configured similarly to the CNN for consistency in performance evaluation. This model was optimized using the RMSprop optimizer, with a learning rate of 0.001, chosen for its effectiveness in maintaining a stable convergence on smaller networks. Training was conducted over 50 epochs for the SNN and 100 epochs for the CNN. Batch sizes were set at 128 for the CNN and 64 for the SNN to balance memory constraints and training dynamics. The loss function across both models was categorical cross-entropy, and performance metrics included accuracy and F1 score to evaluate both precision and recall effectively.

B. Evaluation Metrics

Model Accuracy: We evaluate the model's performance through Test Accuracy on the dataset $\mathcal{D}_u$ from which specific data has been unlearned. This metric provides insight

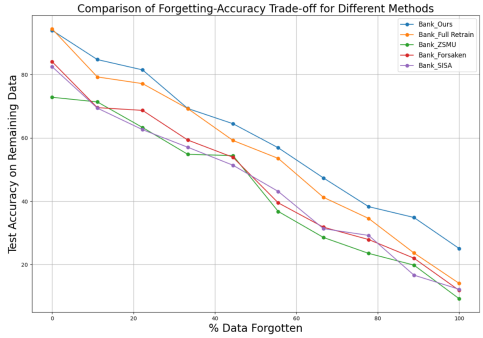


Fig. 5. Forgetting-accuracy trade-off of non_IID and CNN of credit.

into how well the model retains its predictive capabilities post-unlearning. We also report the model's test accuracy on $\mathcal{D}_r$.

Membership Inference Attack (MIA): To evaluate unlearning effectiveness, we use MIA, a standard method for assessing neural network privacy [51], [52]. The attack model, $\mathcal{F}(.)$, identifies if a sample was in the training set. The Forget Rate is defined as $1 - \mathbb{P}[\mathcal{F}(Y_t) = \kappa]$, where Y_t is the unlearned model's output, and κ is the probability of being recognized as training data. Lower Forget Rates indicate better unlearning.

Unlearn Time and Relearn Time: Unlearn Time measures epochs required to achieve a 20% Forget Rate. Relearn Time quantifies epochs needed to restore 95% of the initial test accuracy on $\mathcal{D}_u$ after unlearning.

Forgetting-Accuracy Trade-off: This metric evaluates the balance between forgetting specific data and maintaining accuracy. It compares accuracy on the full dataset $\mathcal{D}$ and the subset $\mathcal{D}\backslash\mathcal{D}_u$, reflecting the model's utility post-unlearning.

C. Results

Table II provides a detailed overview of the performance metrics for machine unlearning methods applied to two datasets, "Bank" and "Credit", using different models under IID and Non-IID distributions. For each combination of dataset, model, and data distribution, the table lists six key metrics: Accuracy on the dataset $\mathcal{D}_r$, F1 Score on $\mathcal{D}_r$, Accuracy on $\mathcal{D}_u$, Forget Rate, Unlearn Time (measured in epochs), and Relearn Time (also in epochs). The accuracy metrics indicate how well the models perform on the remaining data after unlearning ($\mathcal{D}_r$) and on the data that should be forgotten ($\mathcal{D}_u$), while the F1 score provides a measure of the model's precision and recall.

From the table, it can be observed that the models generally maintain high accuracy and F1 scores on $\mathcal{D}_r$, suggesting that they effectively retain their predictive capabilities on the remaining data post-unlearning. The Accuracy on $\mathcal{D}_u$ is notably lower, which aligns with the intended effect of unlearning, indicating reduced model performance on the data that should be forgotten. The Forget Rate varies slightly, with higher rates indicating more effective unlearning. Unlearn Times and Relearn Times are variable, reflecting the complexity and computational demand of unlearning and relearning processes across different model setups. Notably, models on the Credit dataset generally require

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

TABLE II
UNLEARNING PERFORMANCE OF OUR METHOD UNDER DIFFERENT SETTINGS

Dataset	Distribution/Model	Accuracy on $\mathcal{D}_r$	F1 Score on $\mathcal{D}_r$	Accuracy on $\mathcal{D}_u$	Forget Rate	Unlearn Time	Relearn Time
Bank	IID/CNN	94.5%	0.91	52.1%	0.85	50	205
Bank	IID/SNN	90.3%	0.89	48.7%	0.80	32	159
Bank	Non-IID/CNN	93.2%	0.92	51.4%	0.88	62	252
Bank	Non-IID/SNN	89.1%	0.91	47.0%	0.83	49	185
Credit	IID/CNN	96.2%	0.92	54.5%	0.90	83	306
Credit	IID/SNN	91.8%	0.90	49.9%	0.82	54	203
Credit	Non-IID/CNN	95.0%	0.93	43.3%	0.87	72	341
Credit	Non-IID/SNN	90.5%	0.91	48.7%	0.85	64	222

TABLE III
COMPARISON OF UNLEARNING PERFORMANCE OF DIFFERENT METHODS

Dataset	Distribution / Model	Ours			Full Retrain			ZSMU			Forsaken			SISA		
		Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR
Bank	IID/CNN	**95%**	**45%**	**0.90**	96%	41%	0.87	92%	50%	0.85	90%	55%	0.80	93%	47%	0.88
Bank	IID/SNN	**92%**	**48%**	**0.85**	93%	42%	0.82	89%	53%	0.78	88%	58%	0.75	90%	50%	0.83
Bank	Non-IID/CNN	**94%**	**42%**	**0.88**	95%	36%	0.85	91%	49%	0.83	89%	54%	0.77	92%	44%	0.86
Bank	Non-IID/SNN	**91%**	**46%**	**0.82**	92%	40%	0.79	88%	52%	0.76	86%	57%	0.73	89%	48%	0.80
Credit	IID/CNN	**96%**	**40%**	**0.92**	97%	33%	0.89	93%	45%	0.87	91%	50%	0.82	94%	42%	0.90
Credit	IID/SNN	**93%**	**43%**	**0.87**	94%	37%	0.84	90%	48%	0.80	89%	53%	0.77	91%	45%	0.85
Credit	Non-IID/CNN	**95%**	**38%**	**0.90**	96%	31%	0.87	92%	44%	0.85	90%	49%	0.79	93%	39%	0.88
Credit	Non-IID/SNN	**92%**	**41%**	**0.85**	93%	35%	0.82	89%	47%	0.78	87%	52%	0.74	90%	43%	0.83

Acc_r, Acc_u, and FR denote Accuracy on $\mathcal{D}_r$, Accuracy on $\mathcal{D}_u$, and Forget Rate, respectively.

more epochs for both unlearning and relearning, which could suggest a higher complexity or a larger scale of data compared to the Bank dataset.

We compare our proposal with several machine unlearning solutions. In particular, we consider Zero-Shot Full Retraining, Zero-Shot Machine Unlearning (ZSMU) [53], Forsaken [54], and SISA [36] as our baselines. The results is shown in Table III. These baselines were chosen for their focus on different aspects of the unlearning process: ZSMU emphasizes zero-shot unlearning, Forsaken targets secure data removal, and SISA optimizes the computational efficiency of the unlearning process. To comprehensively evaluate our method, we used accuracy, forget rate, and unlearn time as metrics, which align with the study's objective of balancing privacy protection, model performance, and computational efficiency. The performance metrics include Accuracy on $\mathcal{D}_r$ (Acc_r), which gauges the model's accuracy on the remaining data after unlearning; Accuracy on $\mathcal{D}_u$ (Acc_u), which measures accuracy on the data that should have been forgotten, and the Forget Rate (FR), a crucial indicator of the effectiveness of the unlearning process.

The results illustrate varying degrees of success for each method in achieving high Acc_r and low Acc_u, which indicates effective unlearning. Our proposal generally provides a balanced performance with high Acc_r and moderate to low Acc_u, coupled with high FR, suggesting robust unlearning capability. The "Full Retrain" method, while sometimes offering the highest Acc_r, does not always result in the lowest Acc_u, indicating less effective unlearning compared to our proposal. The methods ZSMU, Forsaken, and SISA show a diverse range of outcomes with ZSMU generally maintaining higher Acc_u, potentially indicating weaker unlearning. Forsaken and SISA methods demonstrate variable but often effective unlearning capabilities, with SISA occasionally outperforming others in terms of FR. This data is critical for understanding the trade-offs between maintaining model performance and effectively erasing unwanted data.

Figs. 2 and 3 offer a detailed analysis of the Forgetting-Accuracy Trade-off for different model settings within banking and credit data scenarios, presenting a compelling visual depiction of how different machine learning models perform under various data forgetting percentages. In Fig. 2, the focus is on the Bank dataset, where the performance of both SNN and CNN is analyzed across two types of data distributions. This figure distinctly showcases that CNN models, particularly in non-IID configurations, maintain higher accuracy at lower levels of data forgetting, underscoring their robustness in handling more complex, varied data patterns that do not follow identical probability distributions. This suggests a stronger resilience of CNNs against data removal, potentially due to their deep, layered structure which might capture intrinsic data features more effectively than SNNs.

Fig. 3 extends this analysis to the credit dataset, again comparing the performances of SNN and CNN models under similar IID and non-IID conditions. In both datasets, CNN models outperform SNNs, as evidenced by their slower decline in accuracy as the percentage of data forgotten increases. This indicates that CNNs can better generalize from remaining data, potentially reducing the impact of data forgetting. The results also highlight the importance of choosing the right model architecture and considering the nature of data distribution when designing systems for sectors where data deletion and privacy are periodically necessary yet challenging to manage without compromising on the system's operational effectiveness.

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF UNLEARNING PERFORMANCE UNDER REMOVING DIFFERENTIAL PRIVACY WITH ADAPTIVE NOISE SCALING (ANS) OR DYNAMIC LEARNING RATE ADJUSTMENT (DLRA)

Dataset	Distribution / Model	ANS+DLRA			DLRA			ANS			DLRA (weighted)			ANS (smaller scale)		
		Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR	Acc_r	Acc_u	FR
Bank	IID/CNN	94%	44%	0.88	93%	42%	0.83	92%	52%	0.84	91%	57%	0.82	94%	49%	0.87
Bank	IID/SNN	93%	46%	0.83	94%	44%	0.85	88%	54%	0.81	89%	60%	0.78	91%	52%	0.82
Bank	Non-IID/CNN	93%	43%	0.89	92%	38%	0.86	92%	50%	0.86	90%	53%	0.79	91%	46%	0.87
Bank	Non-IID/SNN	90%	47%	0.83	91%	39%	0.80	87%	51%	0.77	85%	59%	0.75	88%	47%	0.81
Credit	IID/CNN	95%	39%	0.91	93%	32%	0.86	94%	44%	0.86	92%	49%	0.81	91%	41%	0.89
Credit	IID/SNN	94%	45%	0.88	93%	36%	0.85	91%	49%	0.82	88%	55%	0.76	92%	44%	0.86
Credit	Non-IID/CNN	94%	37%	0.91	95%	33%	0.83	91%	43%	0.84	89%	48%	0.78	92%	40%	0.87
Credit	Non-IID/SNN	91%	42%	0.84	92%	34%	0.83	90%	46%	0.79	88%	51%	0.76	91%	42%	0.84

Acc_r, Acc_u, and FR denote Accuracy on $\mathcal{D}_r$, Accuracy on $\mathcal{D}_u$, and Forget Rate.DLRA (weighted) and ANS (smaller scale) denote multiplying 0.5 to the adjusted learning rate and noise added to the learning process, respectively.

TABLE V
DETAILED UNLEARNING PERFORMANCE

Dataset	Distribution/Model	Accuracy on D_r	F1 Score on D_r	Accuracy on D_u	Forget Rate	Unlearn Time (epochs)	Relearn Time (epochs)
Bank	Skewed/CNN	92.7%	0.89	56.3%	0.80	45	195
Bank	Skewed/SNN	88.5%	0.86	53.7%	0.76	30	140
Bank	IID+Noise/CNN	93.9%	0.91	50.5%	0.85	65	230
Bank	IID+Noise/SNN	90.0%	0.88	48.2%	0.82	40	180
Credit	Non-IID/Hetero-CNN	94.3%	0.92	54.0%	0.87	70	310
Credit	Non-IID/Hetero-SNN	89.8%	0.89	50.8%	0.82	50	220
Credit	Overlapping/CNN	95.5%	0.93	49.0%	0.88	78	345
Credit	Overlapping/SNN	92.1%	0.91	44.7%	0.84	55	265

Figs. 4 and 5 depict the Forgetting-Accuracy Trade-off for different methods applied to non-IID and CNN. Each figure illustrates the test accuracy on remaining data as the percentage of data forgotten increases, represented by multiple lines each corresponding to a different method of data forgetting. Specifically, the methods shown include our proposal, Full Retrain, ZSMU, Forsaken, and SISA, allowing for a comparative analysis of how each technique impacts the model's ability to maintain accuracy. The lines exhibit a general downward trend, highlighting a decrease in accuracy as more data is removed, with variations in the rate of decline among different methods, suggesting that some approaches are more effective at preserving accuracy despite data reduction. In both figures, the performance of our proposal typically shows competitive or superior retention of accuracy compared to other methods, particularly against the more drastic declines seen in methods like Full Retrain and SISA. These results are crucial for understanding the effectiveness of various unlearning techniques in practical scenarios. This figure clearly demonstrate the trade-offs involved in unlearning, where maintaining high accuracy levels while ensuring compliance presents a significant challenge.

Table V presents the detailed unlearning performance. Various data distributions such as Skewed, IID+Noise, Non-IID/Hetero are considered. The results show that CNN models generally achieve higher accuracy and F1 scores on the remaining data (D_r), while also maintaining effective forget rates and efficient unlearning times across different settings. The Skewed and IID+Noise setups exhibit decent performance, with forget rates ranging from 0.76 to 0.85. Non-IID/Hetero settings show robustness, particularly with the CNN models reaching a forget rate of 0.87. The Overlapping distribution shows that CNN models can maintain a high accuracy (95.5%) even when data is mixed across clients, though at the cost of increased unlearn and relearn times, indicating a trade-off between data complexity and model efficiency. Overall, the models demonstrate effective unlearning across diverse scenarios, with varying degrees of efficiency based on the data distribution and model type.

D. Ablation Study

In this subsection, we investigate two algorithms involved in our proposal. Table IV presents a comparison of unlearning performance on removing Differential Privacy with Adaptive Noise Scaling (ANS) or Dynamic Learning Rate Adjustment (DLRA). The experiments measure the effectiveness of these approaches across multiple datasets (Bank and Credit) and models (IID/CNN, IID/SNN, Non-IID/CNN, Non-IID/SNN) in terms of three key metrics: Accuracy on remaining data, Accuracy on forgotten data and Forget Rate (FR). The table includes results from four distinct methodologies: ANS+DLRA, DLRA alone, ANS, and DLRA when applied on a smaller scale or with a weighted adjustment.

From the results, it is evident that the combination of ANS and DLRA (ANS+DLRA) generally leads to a balanced performance in maintaining accuracy on the remaining data while achieving considerable forgetting rates. In contrast, applying DLRA alone tends to result in lower accuracy on both remaining and forgotten data, indicating a potential compromise in model performance without the adaptive noise scaling. The column labeled ANS shows a slightly better performance in terms of accuracy on forgotten data compared to DLRA, suggesting

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF THE REQUIRED UNLEARNING EPOCHS TO REACH A CERTAIN FORGET RATE FOR DIFFERENT UNLEARNING METHODS

Dataset	Distribution/Model	ZSMU	Forsaken	SISA	Ours
Bank	IID/CNN	70	65	60	48
Bank	IID/SNN	55	50	45	30
Bank	Non-IID/CNN	90	80	75	60
Bank	Non-IID/SNN	70	65	60	47
Credit	IID/CNN	110	100	90	80
Credit	IID/SNN	80	70	65	52
Credit	Non-IID/CNN	100	90	85	70
Credit	Non-IID/SNN	80	75	70	62

TABLE VII
COMPARISON OF THE REQUIRED UNLEARNING EPOCHS TO REACH A CERTAIN FORGET RATE UNDER DIFFERENT OPTIMIZATION METHODS

Dataset	Distribution/Model	Ours	no **SVRG**	no **QA**	no **PP**
Bank	IID/CNN	48	69	62	52
Bank	IID/SNN	30	55	43	43
Bank	Non-IID/CNN	60	76	71	72
Bank	Non-IID/SNN	47	69	67	59
Credit	IID/CNN	85	89	92	85
Credit	IID/SNN	52	75	76	78
Credit	Non-IID/CNN	60	85	81	81
Credit	Non-IID/SNN	47	82	92	77

QA and PP denote quantum annealing and parallel processing, respectively.

that noise addition helps mitigate the impact of learning rate adjustments alone. Interestingly, when DLRA is applied on a smaller scale or weighted, there is a notable decline in forgetting effectiveness, as shown by generally higher accuracy on forgotten data and lower forget rates. These findings underscore the nuanced trade-offs between different unlearning methods, highlighting the need to tailor privacy-enhancing technologies to specific datasets and model configurations to optimize both privacy and utility.

E. Unlearning Efficiency

Table VI provides a comparison of the required unlearning epochs to reach a certain Forget rate for different methods across various datasets and models. In particular, the unlearning epochs reported for "ZSMU" is measured by finishing the training of the student model. For the Bank dataset using an IID/CNN model, "Ours" requires the fewest epochs (48), showing more efficiency than ZSMU (70), Forsaken (65), and SISA (60). This trend continues across other models like IID/SNN, Non-IID/CNN, and Non-IID/SNN within the Bank dataset, where our proposal consistently requires fewer epochs, demonstrating its superior efficiency in facilitating unlearning. Similarly, for the Credit dataset, our method again outperforms the other methods in both IID and Non-IID distributions for CNN and SNN models, with fewer epochs needed to achieve the Forget rate, highlighting the method's effectiveness across different types of data and model configurations.

Table VII compares the number of epochs required to reach a specified forget rate across different optimization methods. The methods evaluated include our proposal with all optimization techniques and three alternatives where specific optimizations are not used: no Stochastic Variance Reduced Gradient (SVRG), no Quantum Annealing (QA), and no Parallel Processing (PP). Our method consistently requires fewer epochs across almost all configurations compared to the other methods, indicating its efficiency in achieving the desired forget rate. Notably, "no QA" tends to require the most epochs in several configurations, suggesting it is less efficient relative to other methods in those cases.

Our proposed framework ensures resilience against adversarial and inference attacks through a blend of adaptive perturbation, structured optimization, and feature-level control. Adaptive noise scaling and dynamic gradient adjustments mitigate adversarial impacts by preventing small perturbations from altering model behavior, while kernel-based regularization and quantum optimization maintain robustness by controlling feature influence. For inference attacks, differential privacy mechanisms introduce noise to obscure specific data contributions, and feature-based unlearning selectively removes data, reducing residual information leakage. Together, these elements enhance model integrity and privacy, even under adversarial conditions.

VIII. CONCLUSION

This paper presents a robust methodology for feature-based machine unlearning in vertical Federated Learning systems within IoT environments, addressing the crucial need for efficient and privacy-compliant data deletion capabilities. The proposed approach in this paper allows for the selective removal of data influences without necessitating complete model retraining. The extensive experimental evaluations across various IoT datasets demonstrate that our framework not only adheres to stringent privacy regulations but also retains the utility and accuracy of the learning models. Our methodology significantly reduces computational overhead and enhances the scalability and robustness of the unlearning process, making it particularly viable for dynamic and resource-constrained IoT settings. Future work will focus on refining these techniques and exploring their applicability in other complex federated environments to further bolster privacy without compromising on performance.

REFERENCES

[1] L. Yang et al., "A survey on vertical federated learning: From a layered perspective," 2023, *arXiv:2304.01829*, doi: 10.48550/arXiv.2304.01829.
[2] L. Yu et al., "A survey of privacy threats and defense in vertical federated learning: From model life cycle perspective," 2024, *arXiv:2402.03688*, doi: 10.48550/arXiv.2402.03688.
[3] Y. Liu et al., "A communication efficient vertical federated learning framework," 2019, *arXiv: 1912.11187*. [Online]. Available: http://arxiv.org/abs/1912.11187
[4] K. Yang, T. Fan, T. Chen, Y. Shi, and Q. Yang, "A Quasi-Newton method based vertical federated learning framework for logistic regression," 2019, *arXiv: 1912.00513*. [Online]. Available: http://arxiv.org/abs/1912.00513
[5] S. Yang, B. Ren, X. Zhou, and L. Liu, "Parallel distributed logistic regression for vertical federated learning without third-party coordinator," 2019, *arXiv: 1911.09824*. [Online]. Available: http://arxiv.org/abs/1911.09824
[6] S. Hardy et al., "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," 2017, *arXiv: 1711.10677*. [Online]. Available: http://arxiv.org/abs/1711.10677

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

[7] D. Lia and M. Togan, "Privacy-preserving machine learning using federated learning and secure aggregation," in *Proc. 12th Int. Conf. Electron. Comput. Artif. Intell.*, Bucharest, Romania, 2020, pp. 1–6, doi: 10.1109/ECAI50035.2020.9223127.

[8] D. Wu, M. Pan, Z. Xu, Y. Zhang, and Z. Han, "Towards efficient secure aggregation for model update in federated learning," in *Proc. IEEE Glob. Commun. Conf.*, Taiwan, 2020, pp. 1–6, doi: 10.1109/GLOBECOM42002.2020.9347960.

[9] J. So, B. Güler, and A. S. Avestimehr, "Turbo-aggregate: Breaking the quadratic aggregation barrier in secure federated learning," *IEEE J. Sel. Areas Inf. Theory*, vol. 2, no. 1, pp. 479–489, 2021.

[10] K. A. Bonawitz et al., "Practical secure aggregation for federated learning on user-held data," 2016, *arXiv:1611.04482*. [Online]. Available: http://arxiv.org/abs/1611.04482

[11] P. Schoppmann, A. Gascón, and B. Balle, "Private nearest neighbors classification in federated databases," *IACR Cryptol. ePrint Arch.*, pp. 289, vol. 2018, 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:4894435

[12] G. Wang, B. Gu, Q. Zhang, X. Li, B. Wang, and C. X. Ling, "A unified solution for privacy and communication efficiency in vertical federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, New Orleans, LA, USA, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023, Art. no. 593. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/2b5af479527167d4af78847a9b9b645f-Abstract-Conference.html

[13] R. Wu et al., "Privacy-preserving design of graph neural networks with applications to vertical federated learning," 2023, *arXiv:2310.20552*, doi: 10.48550/arXiv.2310.20552.

[14] S. Wan et al., "How global observation works in federated learning: Integrating vertical training into horizontal federated learning," *IEEE Internet Things J.*, vol. 10, no. 11, pp. 9482–9497, Jun. 2023, doi: 10.1109/JIOT.2023.3234196.

[15] R. Yang, J. Ma, J. Zhang, S. Kumari, S. Kumar, and J. J. P. C. Rodrigues, "Practical feature inference attack in vertical federated learning during prediction in artificial Internet of Things," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 5–16, Jan. 2024, doi: 10.1109/JIOT.2023.3275161.

[16] T. Dimlioglu and A. Choromanska, "GRAWA: Gradient-based weighted averaging for distributed training of deep learning models," in *Proc. Int. Conf. Artif. Intell. Statist.*, PMLR, 2024, pp. 2251–2259.

[17] Y. Luo, Z. Lu, X. Yin, S. Lu, and Y. Weng, "Application research of vertical federated learning technology in banking risk control model strategy," in *Proc. IEEE Int. Conf. Parallel Distrib. Process. Appl., Big Data Cloud Comput., Sustain. Comput. Commun., Social Comput. Netw.*, Wuhan, China, 2023, pp. 545–552, doi: 10.1109/ISPA-BDCloud-SocialCom-SustainCom59178.2023.00103.

[18] E. Hu, Y. Tang, A. Kyrillidis, and C. Jermaine, "Federated learning over images: Vertical decompositions and pre-trained backbones are difficult to beat," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Paris, France, 2023, pp. 19328–19339, doi: 10.1109/ICCV51070.2023.01776.

[19] K. Yang, Z. Song, Y. Zhang, Y. Zhou, X. Sun, and J. Wang, "Model optimization method based on vertical federated learning," in *Proc. IEEE Int. Symp. Circuits Syst.*, Daegu, South Korea, 2021, pp. 1–5, doi: 10.1109/ISCAS51556.2021.9401521.

[20] C. Yu, S. Shen, S. Wang, K. Zhang, and H. Zhao, "Communication-efficient hybrid federated learning for e-health with horizontal and vertical data partitioning," 2024, *arXiv:2404.10110*, doi: 10.48550/arXiv.2404.10110.

[21] Q. Li et al., "Vertical federated learning: Taxonomies, threats, and prospects," 2023, *arXiv:2302.01550*, doi: 10.48550/arXiv.2302.01550.

[22] P. Kairouz et al., "Advances and open problems in federated learning," *Found. Trends Mach. Learn.*, vol. 14, no. 1/2, pp. 1–210, 2021, doi: 10.1561/2200000083.

[23] F. Z. Errounda and Y. Liu, "Adaptive differential privacy in vertical federated learning for mobility forecasting," *Future Gener. Comput. Syst.*, vol. 149, pp. 531–546, 2023, doi: 10.1016/j.future.2023.07.033.

[24] A. Li et al., "Efficient and privacy-preserving feature importance-based vertical federated learning," *IEEE Trans. Mobile Comput.*, vol. 23, no. 6, pp. 7238–7255, Jun. 2024, doi: 10.1109/TMC.2023.3333879.

[25] Y. Zheng, S. Xu, S. Wang, Y. Gao, and Z. Hua, "Privet: A privacy-preserving vertical federated learning service for gradient boosted decision tables," *IEEE Trans. Serv. Comput.*, vol. 16, no. 5, pp. 3604–3620, Sep./Oct. 2023, doi: 10.1109/TSC.2023.3279839.

[26] H. Gu, J. Luo, Y. Kang, L. Fan, and Q. Yang, "FedPass: Privacy-preserving vertical federated deep learning with adaptive obfuscation," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, Macao, SAR, China, 2023, pp. 3759–3767, doi: 10.24963/ijcai.2023/418.

[27] R. Xu, N. Baracaldo, Y. Zhou, A. Anwar, J. Joshi, and H. Ludwig, "FedV: Privacy-preserving federated learning over vertically partitioned data," in *Proc. 14th ACM Workshop Artif. Intell. Secur.*, N. Carlini, A. Demontis, and Y. Chen, Eds., Republic of Korea, 2021, pp. 181–192, doi: 10.1145/3474369.3486872.

[28] J. Sun et al., "Communication-efficient vertical federated learning with limited overlapping samples," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Paris, France, 2023, pp. 5180–5189, doi: 10.1109/ICCV51070.2023.00480.

[29] H. Xu, T. Zhu, L. Zhang, W. Zhou, and P. S. Yu, "Machine unlearning: A survey," *ACM Comput. Surv.*, vol. 56, no. 1, pp. 9:1–9:36, 2024, doi: 10.1145/3603620.

[30] M. Kurmanji, E. Triantafillou, and P. Triantafillou, "Machine unlearning in learned databases: An experimental analysis," *Proc. ACM Manag. Data*, vol. 2, no. 1, pp. 49:1–49:26, 2024, doi: 10.1145/3639304.

[31] H. Lee, Q. Zhang, C. Yang, J. Lou, and L. Xiong, "Contrastive unlearning: A contrastive approach to machine unlearning," 2024, *arXiv:2401.10458*, doi: 10.48550/arXiv.2401.10458.

[32] B. A. Juliussen, J. P. Rui, and D. Johansen, "Algorithms that forget: Machine unlearning and the right to erasure," *Comput. Law Secur. Rev.*, vol. 51, 2023, Art. no. 105885, doi: 10.1016/j.clsr.2023.105885.

[33] Y. Cao and J. Yang, "Towards making systems forget with machine unlearning," in *Proc. 2015 IEEE Symp. Secur. Privacy*, San Jose, CA, USA, 2015, pp. 463–480, doi: 10.1109/SP.2015.35.

[34] M. Chen, Z. Zhang, T. Wang, M. Backes, M. Humbert, and Y. Zhang, "When machine unlearning jeopardizes privacy," in *Proc. 2021 ACM SIGSAC Conf. Comput. Commun. Secur.*, Y. Kim, J. Kim, G. Vigna, and E. Shi, Eds., Virtual Event, Republic of Korea, 2021, pp. 896–911, doi: 10.1145/3460120.3484756.

[35] E. Ullah, T. Mai, A. Rao, R. A. Rossi, and R. Arora, "Machine unlearning via algorithmic stability," in *Proc. Conf. Learn. Theory*, M. Belkin and S. Kpotufe, Eds., Boulder, Colorado, USA, PMLR, 2021, pp. 4126–4142. [Online]. Available: http://proceedings.mlr.press/v134/ullah21a.html

[36] L. Bourtoule et al., "Machine unlearning," in *Proc. 42nd IEEE Symp. Secur. Privacy*, San Francisco, CA, USA, 2021, pp. 141–159, doi: 10.1109/SP40001.2021.00019.

[37] C. Pan, J. Sima, S. Prakash, V. Rana, and O. Milenkovic, "Machine unlearning of federated clusters," in *Proc. 11th Int. Conf. Learn. Representations*, Kigali, Rwanda, 2023. [Online]. Available: https://openreview.net/pdf?id=VzwfoFyYDga

[38] S. Asthana, B. Zhang, R. Mahindru, I. S. Banipal, and P. Chowdhary, "IDMU: Impact driven machine unlearning," in *Proc. IEEE Int. Conf. Big Data*, J. He, T. Palpanas, X. Hu, A. Cuzzocrea, D. Dou, D. Slezak, W. Wang, A. Gruca, J. C. Lin, and R. Agrawal, Eds., Sorrento, Italy, 2023, pp. 5146–5155, doi: 10.1109/BigData59044.2023.10386841.

[39] L. Zhang, T. Zhu, H. Zhang, P. Xiong, and W. Zhou, "FedRecovery: Differentially private machine unlearning for federated learning frameworks," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 4732–4746, 2023, doi: 10.1109/TIFS.2023.3297905.

[40] R. Chen et al., "Fast model DeBias with machine unlearning," in *Proc. Adv. Neural Inf. Process. Syst.*, New Orleans, LA, USA, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023, Art. no. 639. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/2ecc80084c96cc25b11b0ab995c25f47-Abstract-Conference.html

[41] M. Kurmanji, P. Triantafillou, J. Hayes, and E. Triantafillou, "Towards unbounded machine unlearning," in *Proc. Adv. Neural Inf. Process. Syst.*, New Orleans, LA, USA, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023, Art. no. 95. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/062d711fb777322e2152435459e6e9d9-Abstract-Conference.html

[42] J. Jia et al., "Model sparsity can simplify machine unlearning," in *Proc. Adv. Neural Inf. Process. Syst.*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., New Orleans, LA, USA, 2023, Art. no. 2246. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/a204aa68ab4e970e1ceccfb5b5cdc5e4-Abstract-Conference.html

[43] C. Wang, Q. Li, Z. Xiang, and D. Wang, "Has approximate machine unlearning been evaluated properly? From auditing to side effects," 2024, *arXiv:2403.12830*, doi: 10.48550/arXiv.2403.12830.

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.

[44] A. Thudi, H. Jia, I. Shumailov, and N. Papernot, "On the necessity of auditable algorithmic definitions for machine unlearning," in *Proc. 31st USENIX Secur. Symp.*, K. R. B. Butler and K. Thomas, Eds., Boston, MA, USA, USENIX Association, 2022, pp. 4007–4022. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/thudi

[45] W. Wang, C. Zhang, Z. Tian, and S. Yu, "Machine unlearning via representation forgetting with parameter self-sharing," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 1099–1111, 2024, doi: 10.1109/TIFS.2023.3331239.

[46] H. Zhang, T. Nakamura, T. Isohara, and K. Sakurai, "A review on machine unlearning," *SN Comput. Sci.*, vol. 4, no. 4, 2023, Art. no. 337, doi: 10.1007/s42979-023-01767-4.

[47] J. Fan, K. Wu, Y. Zhou, Z. Zhao, and S. Huang, "Fast model update for IoT traffic anomaly detection with machine unlearning," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8590–8602, May 2023, doi: 10.1109/JIOT.2022.3214840.

[48] K. Tanaka, T. Hori, and H. O. Wang, "A multiple Lyapunov function approach to stabilization of fuzzy control systems," *IEEE Trans. Fuzzy Syst.*, vol. 11, no. 4, pp. 582–589, Aug. 2003, doi: 10.1109/TFUZZ.2003.814861.

[49] S. Moro, P. Cortez, and P. Rita, "A data-driven approach to predict the success of bank telemarketing," *Decis. Support Syst.*, vol. 62, pp. 22–31, 2014, doi: 10.1016/j.dss.2014.03.001.

[50] I. Yeh and C. Lien, "The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients," *Expert Syst. Appl.*, vol. 36, no. 2, pp. 2473–2480, 2009, doi: 10.1016/j.eswa.2007.12.020.

[51] A. Salem, Y. Zhang, M. Humbert, P. Berrang, M. Fritz, and M. Backes, "ML-leaks: Model and data independent membership inference attacks and defenses on machine learning models," in *Proc. 26th Annu. Netw. Distrib. Syst. Secur. Symp.*, San Diego, CA, USA, 2019.

[52] H. Xu, T. Zhu, L. Zhang, W. Zhou, and P. S. Yu, "Machine unlearning: A survey," 2023, *arXiv:2306.03558*, doi: 10.48550/arXiv.2306.03558.

[53] V. S. Chundawat, A. K. Tarun, M. Mandal, and M. S. Kankanhalli, "Zero-shot machine unlearning," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 2345–2354, 2023, doi: 10.1109/TIFS.2023.3265506.

[54] Z. Ma, Y. Liu, X. Liu, J. Liu, J. Ma, and K. Ren, "Learn to forget: Machine unlearning via neuron masking," *IEEE Trans. Dependable Secur. Comput.*, vol. 20, no. 4, pp. 3194–3207, Jul./Aug. 2023, doi: 10.1109/TDSC.2022.3194884.

Authorized licensed use limited to: Bangladesh Univ of Engineering and Technology. Downloaded on January 30,2026 at 13:05:23 UTC from IEEE Xplore. Restrictions apply.